The empirical recurrence for OEIS A235909: recovering a conjecture that is not written down

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Abstract

OEIS A235909 records "Empirical recurrence of order 62", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 71$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 62, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 62 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235909 is "Number of (n+1)X(6+1) 0..2 arrays with the minimum plus the maximum unequal to the lower median plus the upper median in every 2X2 subblock.". It has offset 1 and begins

$$224064, 27447272, 3481713664, 454589648064, 59328151603200, 7811201385507328, 1025006860079382528, 13502$$

The entry states:

> Empirical recurrence of order 62 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 08:55:49, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2187$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 71$ of the 2187, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 71$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 4374$ exact terms of the model returns order 62 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{62} c_i\, a(n-i),$$

c_1	170	c_{17}	21298981592895423415564097290240	c_{33}	
c_2	25936	c_{18}	−2917040636068528482386733443317760	c_{34}	
c_3	−5557728	c_{19}	−4805373073668346770805147643674624	c_{35}	
c_4	−207692352	c_{20}	3052550801768414927670302508213862400	c_{36}	
c_5	75552622592	c_{21}	−14082145540667233186738718031063547904	c_{37}	
c_6	−79892318720	c_{22}	−2380869304203793490187688349559154540544	c_{38}	
c_7	−568995774089216	c_{23}	22683475100025313945076793672590096859136	c_{39}	
c_8	11894599641456640	c_{24}	1379634248678386880352218181798804198522880	c_{40}	
c_9	2673917974404587520	c_{25}	−19007631028766641817373584955950634559864832	c_{41}	
c_{10}	−92060781422931640320	c_{26}	−586151141179810746402658323846498189020495872	c_{42}	
c_{11}	−8325966948793890897920	c_{27}	10529127483864713964393438803701664258820407296	c_{43}	
c_{12}	386578290659164005859328	c_{28}	177653600887928149971936743968200293111392370688	c_{44}	
c_{13}	17602734173372454933626880	c_{29}	−4070665031380397055658469208720026464120937119744	c_{45}	
c_{14}	−1065011596598817956911120384	c_{30}	−36376852728472964420160613773808832761250357182464	c_{46}	
c_{15}	−24921031429316229808902897664	c_{31}	1109262390811778339041592398339201017480863555780608	c_{47}	
c_{16}	2065240577314372231089646206976	c_{32}	4412015280005862299068234957833707462218991875915776	c_{48}	−

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 62, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $62+4$ terms removed returns order 62 as well, so $\deg q_{\min} = 62 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $62+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 62 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A235909.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.